Pooling and spatial information

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One readable pooling and spatial-information record.

Compute summaries,explain collisions and challenge unsupported movement/reconstruction claims.

Keep one habit: W1–W3/W5 → selected new map → project spatial ledger → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Convolution as a local operation. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Learned representations. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Regularisation and early stopping. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- A fictional single-channel feature map has rows [1, 4, 2, 0], [3, 2, 5, 1], [0, 6, 1, 3] and [2, 4, 2, 7]. Use a 2 by 2 window, stride 2, no padding and only complete windows. Coordinates are zero-based.

  The four starts are [0, 0], [0, 2], [2, 0] and [2, 2].

- **Max pooling** keeps the largest value in each window. **Mean pooling** adds the four values and divides by 4. Neither operation learns weights in this lesson.

  These are fixed summaries of the incoming feature map, unlike the trainable kernel parameters introduced last week.

- The max **output** is [[4, 5], [6, 7]], and the mean **output** is [[2.5, 2], [3, 3.25]]. Both have shape 2 by 2. Sixteen values become four, but the four **output**s still preserve the coarse arrangement of the four regions.

Convolution as a local operation

> **My remembered idea, example and correction**

Learned representations

My remembered idea, example and correction

Regularisation and early stopping

My remembered idea, example and correction

Lesson reference for workbook practice

Window start	Values	Maximum	Mean
[0, 0]	1, 4, 3, 2	4	2.5
[0, 2]	2, 0, 5, 1	5	2
[2, 0]	0, 6, 2, 4	6	3
[2, 2]	1, 3, 2, 7	7	3.25

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main map [[1, 4, 2, 0], [3, 2, 5, 1], [0, 6, 1, 3], [2, 4, 2, 7]]. Use 2×2 windows, stride 2, no padding, complete windows only. Starts (0, 0), (0, 2), (2, 0), (2, 2) have sums 10, 8, 12, 13; maxima 4, 5, 6, 7; means 2.5, 2, 3, 3.25. The 2×2 outputs preserve coarse region positions and lose detail within a window. Different positions of one 4 among zeros give max 4 and mean 1.

  Two bottom-position windows with one 6 give max 6 and mean 1.5. Window [[-4, -2], [-7, -3]] has max -2 and mean -4. The 3×3 example with a 99 border reads only [[1, 2], [3, 4]], giving max 4 and mean 2.5. Moving an 8 from column 1 to 2 in a 2×4 zero map changes max [[8, 0]] to [[0, 8]] and mean [[2, 0]] to [[0, 2]]. The explicitly declared left-positive head changes class 1 to 0.

  The separate picture uses only the top two main rows and has 1×2 outputs; swapping 1 and 4 inside its left window leaves both summaries unchanged.

- For the lesson window starting at [2, 2], list the four values, sum, maximum and mean. State the complete pooled **output** shape.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main map [[1, 4, 2, 0], [3, 2, 5, 1], [0, 6, 1, 3], [2, 4, 2, 7]]. Use 2×2 windows, stride 2, no padding, complete windows only. Starts (0, 0), (0, 2), (2, 0), (2, 2) have sums 10, 8, 12, 13; maxima 4, 5, 6, 7; means 2.5, 2, 3, 3.25. The 2×2 outputs preserve coarse region positions and lose detail within a window. Different positions of one 4 among zeros give max 4 and mean 1.

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  The separate picture uses only the top two main rows and has 1×2 outputs; swapping 1 and 4 inside its left window leaves both summaries unchanged.

- Give two different 2 by 2 windows containing exactly one 4 and three zeros that have the same maximum and mean. Explain why either scalar alone cannot identify the position of the 4.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main map [[1, 4, 2, 0], [3, 2, 5, 1], [0, 6, 1, 3], [2, 4, 2, 7]]. Use 2×2 windows, stride 2, no padding, complete windows only. Starts (0, 0), (0, 2), (2, 0), (2, 2) have sums 10, 8, 12, 13; maxima 4, 5, 6, 7; means 2.5, 2, 3, 3.25. The 2×2 outputs preserve coarse region positions and lose detail within a window. Different positions of one 4 among zeros give max 4 and mean 1.

  Two bottom-position windows with one 6 give max 6 and mean 1.5. Window [[-4, -2], [-7, -3]] has max -2 and mean -4. The 3×3 example with a 99 border reads only [[1, 2], [3, 4]], giving max 4 and mean 2.5. Moving an 8 from column 1 to 2 in a 2×4 zero map changes max [[8, 0]] to [[0, 8]] and mean [[2, 0]] to [[0, 2]]. The explicitly declared left-positive head changes class 1 to 0.

  The separate picture uses only the top two main rows and has 1×2 outputs; swapping 1 and 4 inside its left window leaves both summaries unchanged.

- Apply the lesson complete-window rule to [[1, 2, 99], [3, 4, 99], [99, 99, 99]]. Give the **output** shape and both **output**s. Explain why neither **output** includes 99.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main map [[1, 4, 2, 0], [3, 2, 5, 1], [0, 6, 1, 3], [2, 4, 2, 7]]. Use 2×2 windows, stride 2, no padding, complete windows only. Starts (0, 0), (0, 2), (2, 0), (2, 2) have sums 10, 8, 12, 13; maxima 4, 5, 6, 7; means 2.5, 2, 3, 3.25. The 2×2 outputs preserve coarse region positions and lose detail within a window. Different positions of one 4 among zeros give max 4 and mean 1.

  Two bottom-position windows with one 6 give max 6 and mean 1.5. Window [[-4, -2], [-7, -3]] has max -2 and mean -4. The 3×3 example with a 99 border reads only [[1, 2], [3, 4]], giving max 4 and mean 2.5. Moving an 8 from column 1 to 2 in a 2×4 zero map changes max [[8, 0]] to [[0, 8]] and mean [[2, 0]] to [[0, 2]]. The explicitly declared left-positive head changes class 1 to 0.

  The separate picture uses only the top two main rows and has 1×2 outputs; swapping 1 and 4 inside its left window leaves both summaries unchanged.

- A 2 by 4 map is zero except for 8 at row 0 column 1. Move the 8 to column 2 and set its old position to zero. Give the max and mean maps before and after. What claim does this refute?

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use rows [-4, -1, 2, 6], [-3, -2, 4, 0], [5, 1, -8, -2], [3, 7, -4, -6]. Under the lesson rule, give all four sums, the max map and the mean map. Explain why replacing a negative maximum by zero is an error here.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Pool a new map independently

> **My independent explanation, trace and limits**
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MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Add a labelled pooling and **spatial information evidence** sheet to Image-pattern Network Investigation. Show how you can compute max and **mean pooling** under an explicit window rule.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Prove that location can be lost for Pooling and **spatial information**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Pooling and **spatial information**. Proposed explanation: "**Max pooling** turns negative numbers into zero". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use rows [-4, -1, 2, 6], [-3, -2, 4, 0], [5, 1, -8, -2], [3, 7, -4, -6]. Under the lesson rule, give all four sums, the max map and the mean map. Explain why replacing a negative maximum by zero is an error here.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Pool a small map and show the detail it cannot retain.
- Feature map rows [1, 4, 2, 0],[3, 2, 5, 1]. Pool non-overlapping 2×2 windows, stride 2, no padding.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

- Can the max map reveal where the 4 originally appeared? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/22

Pooling and spatial information

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Left window

Values 1, 4, 3, 2: max 4, mean 2.5.

2. Right window

Values 2, 0, 5, 1: max 5, mean 2.

3. Compare summaries

Max map [[4, 5]], mean map [[2.5, 2]], both shape 1 x 2.

CHANGE ONE THING

Swap 1 and 4 within the left window.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.
- For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Pool a small map and show the detail it cannot retain.
- Which labelled step matches this detail? Max map [[4, 5]], mean map [[2.5, 2]], both shape 1×2.
- 1. Left window
- 2. Compare summaries
- 3. Right window
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Compare summaries Max map [[4, 5]], mean map [[2.5, 2]], both shape 1×2. Pooling compresses each local window into a summary.

Maximum keeps its largest value; mean keeps its average. Neither summary records where each value appeared.

Swapping values inside one window leaves both summaries unchanged here, so the pooled **output** cannot reconstruct the original arrangement. The chosen window, stride and boundary policy determine the **output** shape.

Changed-case question and optional hint

- Changed-case practice: Swap 1 and 4 within the left window. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. Both pooled maps remain unchanged.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Both pooled maps remain unchanged. Max and mean ignore the positions of values inside the same window.

Repair and retain the record

Review Convolution as a local operation (ai-5/21). Goal: Compute a local weighted sum at each valid image position. Short practice: Why is flipping the kernel an implementation change here?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Learned representations (ai-5/19). Goal: Trace an **input** through a hidden representation and a decision rule. Short practice: Why can no deterministic head using only the sum classify both correctly?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Regularisation and early stopping (ai-5/16). Goal: Separate a weight penalty from **prediction** error. Short practice: Why should the penalty value not be reported as validation error?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

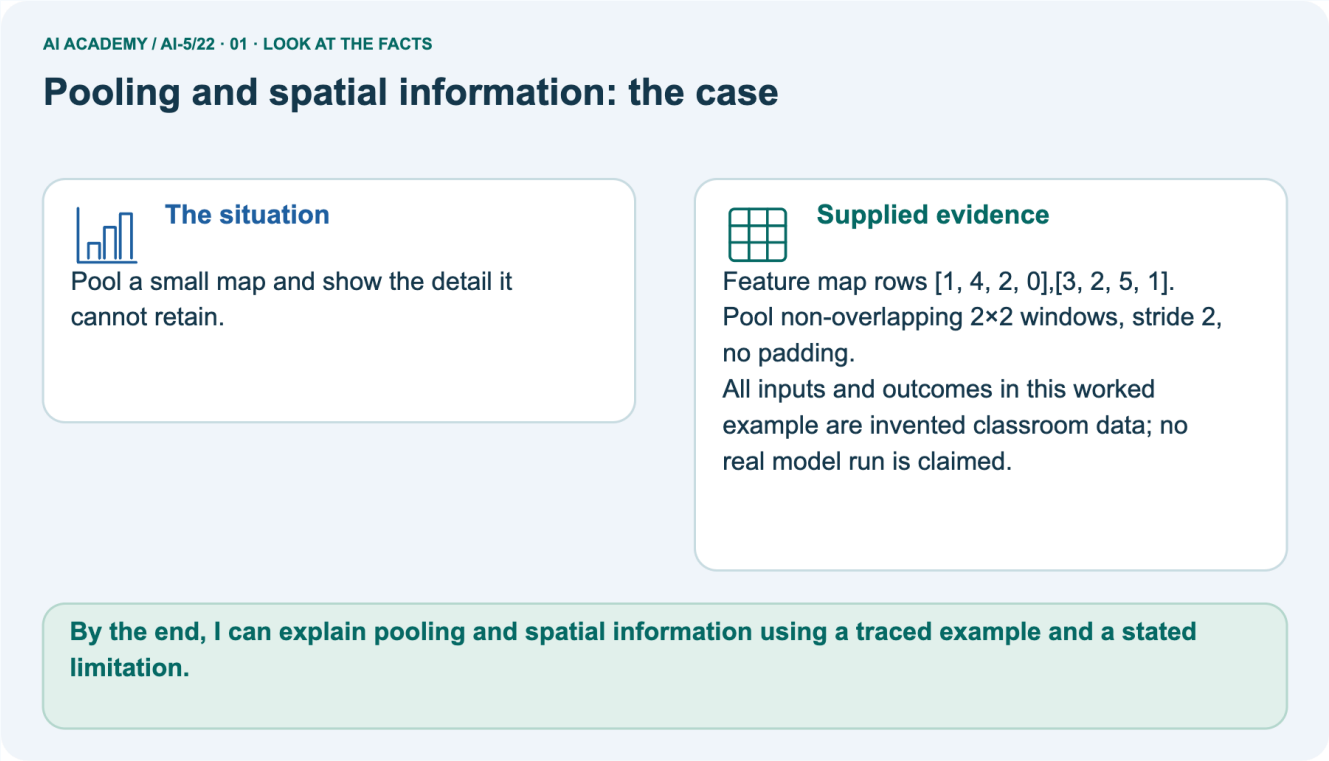
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Pooling and spatial information: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

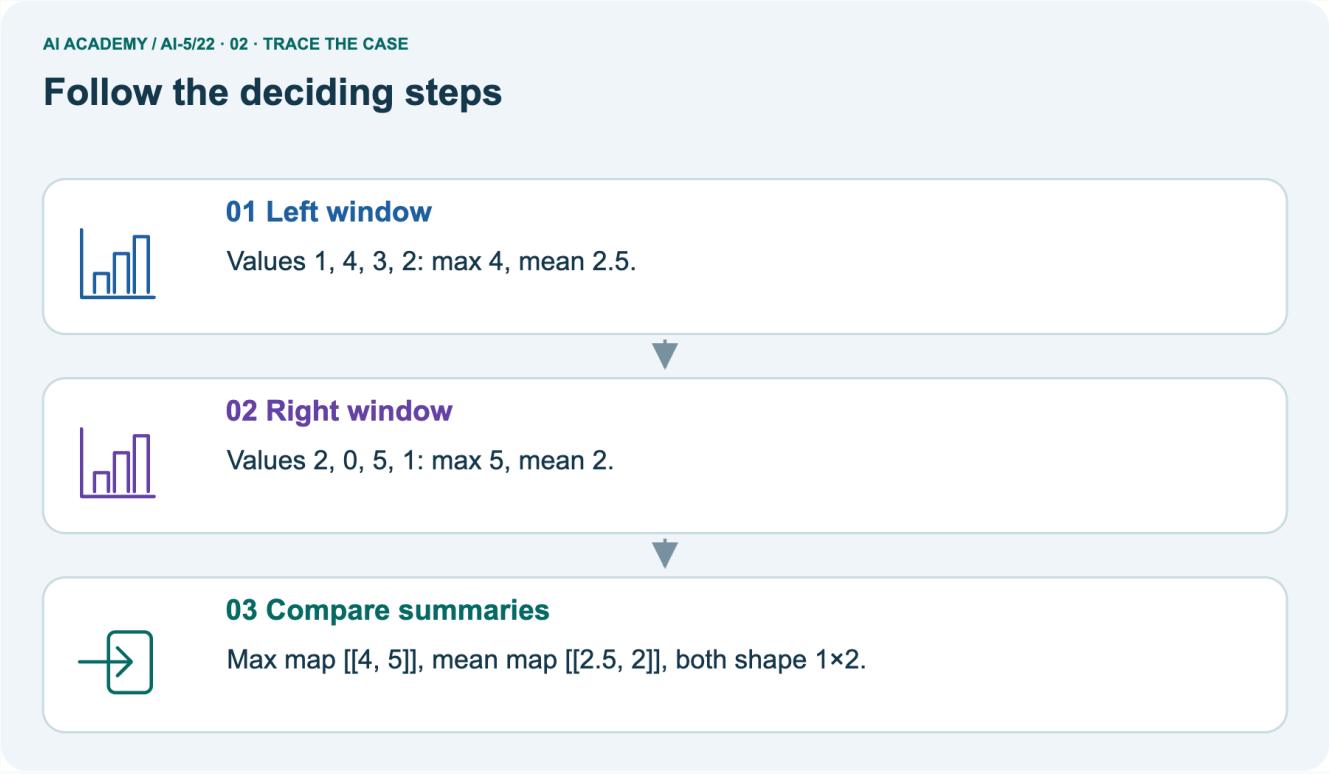
Pool a small map and show the detail it cannot retain. Supplied facts: Feature map rows [1, 4, 2, 0],[3, 2, 5, 1]. Pool non-overlapping 2×2 windows, stride 2, no padding.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Left window: Values 1, 4, 3, 2: max 4, mean 2.5. → Right window: Values 2, 0, 5, 1: max 5, mean 2. → Compare summaries: Max map [[4, 5]], mean map [[2.5, 2]], both shape 1×2.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Two adjacent pooling windows



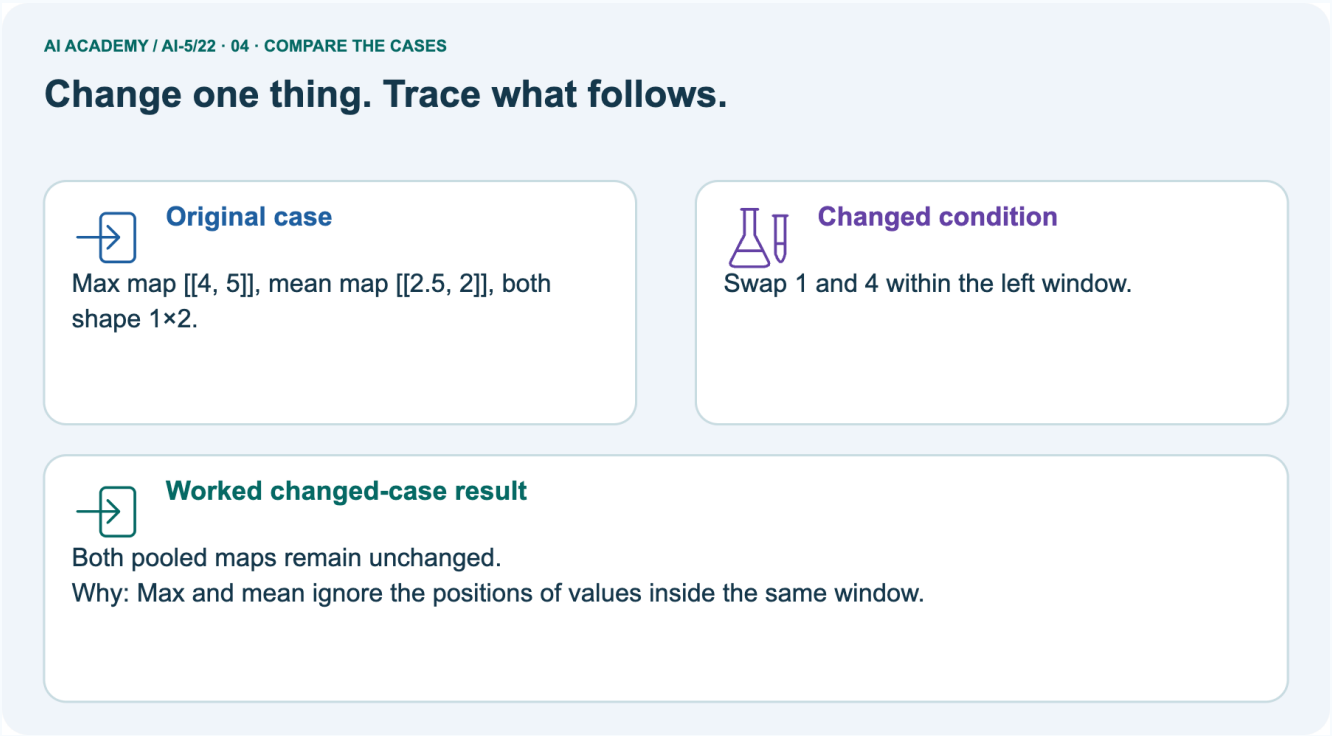
Follow the labelled picture with the matching explanation. Open the vector diagram

Two adjacent pooling windows. row 0: left 0 = 1, left 1 = 4, right 0 = 2, right 1 = 0. row 1: left 0 = 3, left 1 = 2, right 0 = 5, right 1 = 1.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

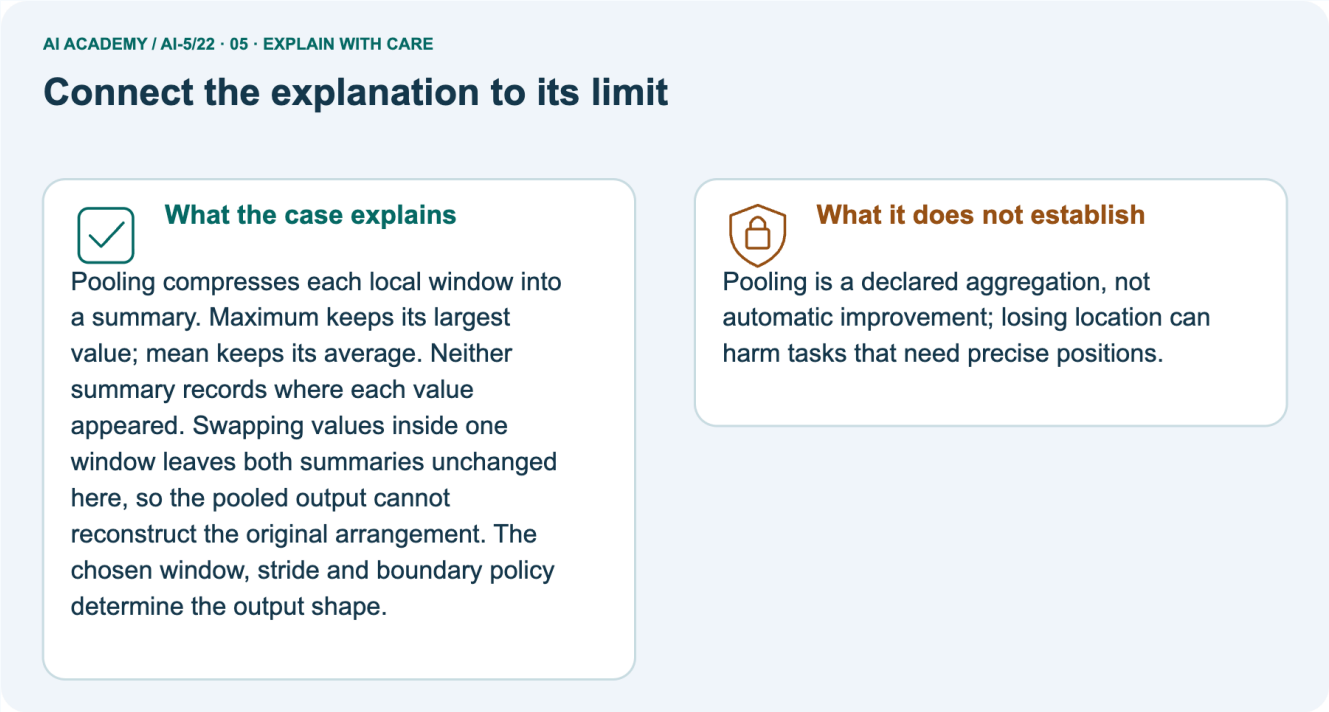
Original case: Max map [[4, 5]], mean map [[2.5, 2]], both shape 1×2. Changed condition: Swap 1 and 4 within the left window. Expected result: Both pooled maps remain unchanged.

Reason: Max and mean ignore the positions of values inside the same window.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Pooling compresses each local window into a summary. Maximum keeps its largest value; mean keeps its average. Neither summary records where each value appeared.

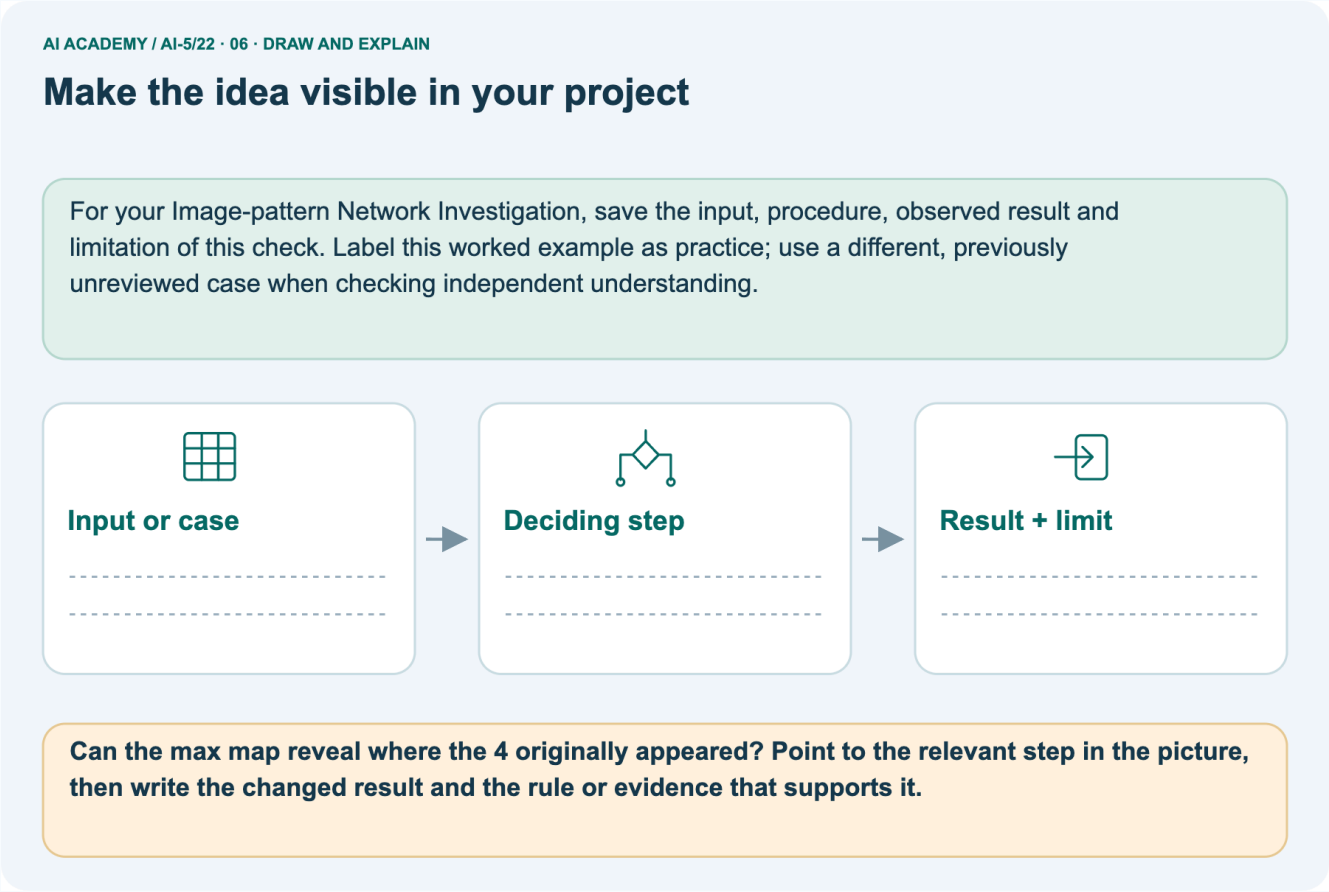
Swapping values inside one window leaves both summaries unchanged here, so the pooled **output** cannot reconstruct the original arrangement. The chosen window, stride and boundary policy determine the **output** shape.

Limit: Pooling is a declared aggregation, not automatic improvement; losing location can harm tasks that need precise positions.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Can the max map reveal where the 4 originally appeared? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

01. Pooling and spatial information: the case

02. Follow the deciding steps

03. Two adjacent pooling windows

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project
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<summary>Optional interactive practice and completed work</summary>

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.
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